

PARIKSHA365 — STUDY NOTES

# Biology — Common Book

SSC, RRB, State PSC, Banks (GA) — General  
Science + Health

SSC

RRB

BANKS

STATE PSC

v 2026-04

**The promise.** Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

# Before You Begin

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Forget memorising isolated facts. Every exam question in biology comes from one of four buckets: **names & discoverers**, **cause & effect** (disease-vector, hormone-disorder), **numbers & extremes** (206 bones, 120-day RBC life), and **comparisons** (prokaryote vs eukaryote, mitosis vs meiosis). This book is structured to hit all four buckets hard, in the order examiners love.

Read end-to-end once for context. Then on your second pass, cover the right column of every table and test yourself. The 25-question mini-mock at the end tells you what you've missed.

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# Examiner Blueprint — What Actually Appears on Papers

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## ⚠ EXAM ALERT

**High-yield order (SSC CGL/CHSL, RRB NTPC, IBPS PO GA section):**

1. **Vitamins & deficiency diseases** — 2-4 questions per paper, reliably. Know every row of the vitamin table cold.
2. **Disease-vector mapping** — Malaria/Dengue/Filaria vectors are tested in every single exam.
3. **Human body extremes** — Smallest/largest bone, organ, gland, cell. Examiners love these.
4. **Hormones & glands** — "Master gland", "fight-or-flight", goitre cause — all predictable.
5. **Cell organelles** — "Powerhouse", "suicide bags", "protein factory".
6. **Blood groups** — Universal donor, universal recipient, discoverer.
7. **Plant biology** — Photosynthesis equation, C3 vs C4, fruit-ripening hormone.
8. **Ecology** — 10% rule, biodiversity hotspots, pyramid of energy (always upright).

**Topics examiners rarely test:** detailed embryology, molecular biology mechanisms, specific enzyme kinetics. Don't over-invest here.

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# PART A — FOUNDATIONS: CLASSIFICATION & THE CELL

## Chapter A1 — Classification of Life

In 1735, a Swedish botanist named Carl Linnaeus decided that humanity needed a universal filing system for every creature on Earth. He invented binomial nomenclature — two-part Latin names — and the hierarchical rank system we still use today.

### The seven taxonomic ranks

Kingdom → Phylum → Class → Order → Family → Genus → Species

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**King Philip Came Over For Great Supper** — Kingdom, Phylum, Class, Order, Family, Genus, Species. This mnemonic appears in textbooks from Class 6 to postgraduate level; it is absolutely exam-safe.

### Whittaker's Five Kingdoms (1969)

| Kingdom  | Members                         | Key feature                   |
|----------|---------------------------------|-------------------------------|
| Monera   | Bacteria, cyanobacteria         | Prokaryotes — no nucleus      |
| Protista | Amoeba, Paramecium, Euglena     | Unicellular eukaryotes        |
| Fungi    | Mushrooms, yeasts, moulds       | Chitin cell wall; decomposers |
| Plantae  | Mosses, ferns, flowering plants | Photosynthetic multicellular  |
| Animalia | Insects, fish, mammals          | Heterotrophic multicellular   |

Later models added Archaea → six kingdoms. Carl Woese proposed three domains: Bacteria, Archaea, Eukarya.

#### ✓ KEY FACT

#### Exam-critical "Fathers" — memorise all five:

Father of taxonomy → **Carl Linnaeus**

Father of modern biology → **Aristotle**

Father of genetics → **Gregor Mendel**

Father of botany → **Theophrastus**

Father of zoology → **Aristotle** (same person)

Binomial name of humans → **Homo sapiens**

## Chapter A2 — The Cell

In 1665, Robert Hooke sliced cork with a primitive microscope and saw small boxy chambers. He called them cells — from the Latin *cellula* meaning "small room". Two centuries later, Schleiden (1838, plants) and Schwann (1839, animals) formalised the cell theory that every living thing is made of cells.

### Three tenets of Cell Theory

1. All living organisms are made of one or more cells.
2. The cell is the basic structural and functional unit of life.
3. New cells arise only from pre-existing cells. (*Virchow, 1855*)

### Prokaryote vs Eukaryote — the fundamental split

| Feature                   | Prokaryote               | Eukaryote                                        |
|---------------------------|--------------------------|--------------------------------------------------|
| True nucleus              | Absent (DNA in nucleoid) | Present                                          |
| Size                      | 1–10 $\mu\text{m}$       | 10–100 $\mu\text{m}$                             |
| Membrane-bound organelles | None                     | Many                                             |
| Ribosomes                 | 70S                      | 80S (cytoplasm); 70S in mitochondria/chloroplast |
| Cell division             | Binary fission           | Mitosis / Meiosis                                |
| Examples                  | Bacteria, cyanobacteria  | Plants, animals, fungi                           |

### Cell organelles — the factory analogy

| Organelle      | Function                   | Memory hook  |
|----------------|----------------------------|--------------|
| <b>Nucleus</b> | DNA storage; controls cell | CEO's office |

| Organelle              | Function                         | Memory hook        |
|------------------------|----------------------------------|--------------------|
| <b>Mitochondria</b>    | ATP (energy) via respiration     | <b>Power plant</b> |
| <b>Ribosome</b>        | Protein synthesis                | 3-D printer        |
| <b>ER (rough)</b>      | Protein folding + transport      | Assembly line      |
| <b>ER (smooth)</b>     | Lipid synthesis, detox           | Pipeline           |
| <b>Golgi apparatus</b> | Packages + dispatches proteins   | Post office        |
| <b>Lysosome</b>        | Digests waste + worn-out parts   | Recycling centre   |
| <b>Chloroplast</b>     | Photosynthesis (plants only)     | Solar panel        |
| <b>Vacuole</b>         | Storage; turgor in plants        | Warehouse          |
| <b>Cell wall</b>       | Rigidity; protection             | Building walls     |
| <b>Centrosome</b>      | Spindle formation (animal cells) | Foreman            |
| <b>Peroxisome</b>      | Breaks $H_2O_2$ via catalase     | Detox unit         |

#### ⚠ EXAM ALERT

##### Four organelle questions examiners love:

"Powerhouse of the cell?" → **Mitochondria**

"Suicide bags?" → **Lysosomes** (contain hydrolytic enzymes; self-digest when cell dies)

"Site of protein synthesis?" → **Ribosomes**

"Which organelle has its own DNA?" → **Mitochondria AND Chloroplast** (both have own circular DNA)

## Cell division — Mitosis vs Meiosis

| Aspect            | Mitosis              | Meiosis                   |
|-------------------|----------------------|---------------------------|
| Location          | Somatic (body) cells | Germ (reproductive) cells |
| Daughter cells    | 2                    | 4                         |
| Chromosome count  | Stays diploid (2n)   | Halved to haploid (n)     |
| Genetic variation | None — clones        | High (crossing over)      |
| Purpose           | Growth, repair       | Making gametes            |
| Phases            | PMAT × 1             | PMAT × 2 (I + II)         |

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**Phases of mitosis: PMAT** — Prophase, Metaphase, Anaphase, Telophase. In meiosis, the same phases happen twice (Meiosis I and Meiosis II), but Prophase I has the special event of crossing over (chiasmata formation), which creates genetic variation.

Body extremes — memorise cold

| Fact                       | Answer                   |
|----------------------------|--------------------------|
| Largest cell (single cell) | Ostrich egg              |
| Smallest cell              | <i>Mycoplasma</i>        |
| Longest cell               | Neuron (axon up to ~1 m) |
| First to see cells         | Robert Hooke (1665)      |
| Discovered nucleus         | Robert Brown (1831)      |

# PART B — GENETICS & EVOLUTION

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## Chapter B1 — Mendelian Genetics

Between 1856 and 1863, Gregor Mendel — an Augustinian friar in Brno, Moravia — grew 28,000 pea plants and discovered that inheritance is particulate, not blended. His paper was published in 1866 and ignored for 34 years. When rediscovered simultaneously by De Vries, Correns, and von Tschermak in 1900, it became the foundation of modern genetics.

### Mendel's Three Laws

1. **Law of Dominance** — In a heterozygote (Tt), the dominant allele (T) masks the recessive allele (t). Only TT and Tt look tall; tt looks short.
2. **Law of Segregation** — Alleles separate during gamete formation. Each gamete gets only one allele from each pair.
3. **Law of Independent Assortment** — Alleles for different traits sort into gametes independently (provided genes are on different chromosomes).

### Monohybrid cross: Tt × Tt

Phenotypic ratio → **3 : 1** (dominant : recessive) Genotypic ratio → **1 TT : 2 Tt : 1 tt**

#### ⚠ EXAM ALERT

**Three questions that appear constantly:**

"Father of genetics?" → **Gregor Mendel**

"Mendel used which plant?" → **Pea plant (Pisum sativum)**

"Classic monohybrid ratio?" → **3:1** (phenotypic) and **1:2:1** (genotypic)

## Chapter B2 — DNA & Molecular Genetics

### The Double Helix

In 1953, James Watson and Francis Crick (using X-ray diffraction data from Rosalind Franklin) published the structure of DNA: a double helix with anti-parallel strands. They received the Nobel Prize in 1962.

Key structural facts: - Two strands run anti-parallel (5'→3' and 3'→5'). - Base pairing: **Adenine–Thymine** (2 hydrogen bonds), **Guanine–Cytosine** (3 hydrogen bonds). - Sugar in DNA = **deoxyribose**; in RNA = **ribose**. - Uracil (U) replaces Thymine (T) in RNA.

### Central Dogma

**DNA → (Transcription) → mRNA → (Translation) → Protein**



Reverse transcription (RNA → DNA) occurs in retroviruses like HIV, via the enzyme reverse transcriptase.

### Three types of RNA

| RNA Type         | Function                                          |
|------------------|---------------------------------------------------|
| mRNA (messenger) | Carries genetic code from nucleus to ribosome     |
| tRNA (transfer)  | Brings amino acids to ribosome during translation |
| rRNA (ribosomal) | Structural component of ribosomes                 |

#### ✓ KEY FACT

##### Genetic code essentials:

Start codon → **AUG** (codes for methionine)

Stop codons → **UAA, UAG, UGA**

Total codons → 64 (for 20 amino acids — degeneracy means multiple codons map to same amino acid)

Human chromosomes → **46 (23 pairs)**; 22 autosome pairs + 1 sex pair (XX female, XY male)

Human Genome Project → 1990–2003

CRISPR-Cas9 Nobel → Doudna + Charpentier (2020)

### Genetic disorders

| Disorder             | Type                | Key fact                                      |
|----------------------|---------------------|-----------------------------------------------|
| Sickle cell anaemia  | Autosomal recessive | Single mutation in haemoglobin $\beta$ -chain |
| Thalassemia          | Autosomal recessive | Reduced globin chain production               |
| Haemophilia          | X-linked recessive  | Defective clotting factor (mainly males)      |
| Colour blindness     | X-linked recessive  | X-linked; mostly males affected               |
| Down syndrome        | Trisomy 21          | Extra chromosome 21                           |
| Turner syndrome      | XO                  | Female with single X chromosome               |
| Klinefelter syndrome | XXY                 | Male with extra X                             |
| Huntington's disease | Autosomal dominant  | CAG repeat expansion; late onset              |

#### × COMMON TRAP

Sex of the baby is determined by the **father's sperm**, not the mother. The mother always contributes X. Father contributes X → daughter; Y → son. Questions asking "who determines sex?" — the answer is the father.

## Chapter B3 — Evolution

Charles Darwin sailed on HMS Beagle (1831–36), observed Galápagos finches with differently shaped beaks for different food sources, and published *On the Origin of Species* in 1859, proposing **natural selection** as the mechanism of evolution.

### Evidence for evolution

| Evidence type          | Example                                                                 |
|------------------------|-------------------------------------------------------------------------|
| Fossils                | Archaeopteryx — link between birds and reptiles                         |
| Homologous organs      | Human arm, whale flipper, bat wing — same structure, different function |
| Analogous organs       | Insect wing, bird wing — same function, different origin                |
| Vestigial organs       | Appendix, coccyx, wisdom teeth                                          |
| Molecular biology      | DNA similarity across species                                           |
| Comparative embryology | Vertebrate embryos look similar in early stages                         |

Human evolution: Australopithecus → Homo habilis → Homo erectus → Homo sapiens (~300,000 years ago, Africa).

#### ✓ KEY FACT

##### Key distinctions examiners test:

Homologous organs → same *origin*, different function (evolution divergence)

Analogous organs → same *function*, different origin (convergent evolution)

Lamarck's theory → inheritance of acquired characters (giraffe neck story) — **disproved**

# PART C — PLANT BIOLOGY

## Chapter C1 — Plant Kingdom Overview

Plants are classified by complexity of structure and reproductive strategy:

| Division     | Examples                      | Key feature                                       |
|--------------|-------------------------------|---------------------------------------------------|
| Thallophyta  | Spirogyra, Chara, Algae       | No roots/stems/leaves                             |
| Bryophyta    | Mosses, Marchantia, Riccia    | "Amphibians of plant kingdom"; no vascular tissue |
| Pteridophyta | Ferns, Equisetum, Selaginella | First vascular plants; no seeds                   |
| Gymnosperms  | Pinus, Cycas, Cedrus          | Naked seeds (no fruit)                            |
| Angiosperms  | All flowering plants          | Seeds enclosed in fruit                           |

### Dicot vs Monocot — always tested

| Feature                  | Dicot                  | Monocot                          |
|--------------------------|------------------------|----------------------------------|
| Seed leaves (cotyledons) | 2                      | 1                                |
| Leaf venation            | Reticulate (net-like)  | Parallel                         |
| Flower parts             | In 4s or 5s            | In 3s                            |
| Root type                | Tap root               | Fibrous                          |
| Examples                 | Bean, mango, rose, pea | Wheat, rice, maize, onion, grass |

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**"One for monocots, two for dicots."** All the world's staple grains (wheat, rice, maize, barley) are monocots. All the world's trees and most garden flowers are dicots. This pattern is an easy reality check.

## Chapter C2 — Photosynthesis

Photosynthesis is how plants manufacture food from air and water using sunlight. The overall equation:



## Two phases

**Light reaction** (in thylakoids): Water is split (photolysis), O<sub>2</sub> is released, and ATP + NADPH are produced.

**Dark reaction / Calvin cycle** (in stroma): ATP + NADPH are used to fix CO<sub>2</sub> into glucose. No light is directly needed, but the products of the light reaction are consumed.

## C3, C4, and CAM plants

| Type | Plants                        | Advantage                                  |
|------|-------------------------------|--------------------------------------------|
| C3   | Wheat, rice, soybean          | Most common; less efficient in heat        |
| C4   | Sugarcane, maize, sorghum     | Efficient in tropical heat and high light  |
| CAM  | Cactus, pineapple, succulents | Open stomata at night; minimise water loss |

### ⚠ EXAM ALERT

#### Top four photosynthesis questions:

"Site of photosynthesis?" → **Chloroplast**

"Main photosynthetic pigment?" → **Chlorophyll-a**

"Where does the Calvin cycle occur?" → **Stroma**

"CAM plants open stomata when?" → **Night** (to reduce water loss in desert conditions)

## Chapter C3 — Plant Hormones

| Hormone                    | Main function                                      | Exam clue                      |
|----------------------------|----------------------------------------------------|--------------------------------|
| <b>Auxin (IAA)</b>         | Cell elongation, phototropism, apical dominance    | Bends towards light            |
| <b>Gibberellin</b>         | Stem elongation, seed germination, breaks dormancy | Makes dwarf plants tall        |
| <b>Cytokinin</b>           | Cell division, delays leaf senescence              | "Stay young" hormone           |
| <b>Abscisic acid (ABA)</b> | Stress response, stomatal closure, seed dormancy   | "Stress hormone"               |
| <b>Ethylene</b>            | Fruit ripening, leaf fall                          | The only gaseous plant hormone |

✓ KEY FACT

"Which hormone ripens fruit?" → **Ethylene** — it is a gas. Commercially, ethylene is used to ripen bananas and mangoes during transport.

"Hormone responsible for apical dominance?" → **Auxin** — suppresses lateral buds; pruning removes the apex and removes auxin dominance, causing bushy growth.

## Chapter C4 — Plant Reproduction

**Asexual:** vegetative propagation (cutting, grafting, layering), spores, micropropagation (tissue culture).

**Sexual:** pollination → fertilisation → seed.

Pollination agents: wind (anemophily), water (hydrophily), insects (entomophily), birds (ornithophily), bats (chiropterophily).

**Double fertilisation** — unique to angiosperms: one sperm fertilises the egg (→ embryo/zygote); a second sperm fertilises the polar nuclei (→ endosperm, the food reserve for the embryo).

### Famous Indian botanists

- **M.S. Swaminathan** — Father of the Indian Green Revolution (high-yield wheat varieties).
  - **Birbal Sahni** — pioneer of palaeobotany in India.
  - **J.C. Bose** — proved that plants respond to stimuli.
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# PART D — HUMAN BODY SYSTEMS

## Chapter D1 — Digestive System

The digestive tract runs about 9 metres from mouth to anus. Each section specialises.

**Pathway:** Mouth → Pharynx → Oesophagus → Stomach → Small intestine (Duodenum → Jejunum → Ileum) → Large intestine (Caecum → Colon → Rectum) → Anus.

### Key enzymes

| Source           | Enzyme                    | Substrate → Product             |
|------------------|---------------------------|---------------------------------|
| Saliva           | Amylase (Ptyalin)         | Starch → Maltose                |
| Stomach          | Pepsin (+ HCl)            | Proteins → Peptones             |
| Pancreas         | Amylase                   | Starch → Maltose                |
| Pancreas         | Lipase                    | Fats → Fatty acids + Glycerol   |
| Pancreas         | Trypsin, Chymotrypsin     | Proteins → Peptides             |
| Liver (via bile) | —                         | Emulsifies fats                 |
| Small intestine  | Maltase, Lactase, Sucrase | Disaccharides → Monosaccharides |

#### ✓ KEY FACT

##### Three digestive anatomy questions examiners love:

"Largest gland in the body?" → **Liver**

"Bile is stored in?" → **Gall bladder**

"Main site of nutrient absorption?" → **Small intestine** (villi and microvilli maximise surface area to ~250 m<sup>2</sup>)

"Stomach acid pH?" → **1.5–3.5**

## Chapter D2 — Respiratory System

**Pathway:** Nose → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli.

- ~300 million alveoli; surface area ~70 m<sup>2</sup>.
- Diaphragm and intercostal muscles drive breathing.
- Normal adult breathing rate: **12–18 per minute**; tidal volume ~500 mL.
- O<sub>2</sub> transported mainly by **haemoglobin** (Hb); CO<sub>2</sub> transported mainly as **bicarbonate (HCO<sub>3</sub><sup>-</sup>)** (~70%).

## Chapter D3 — Circulatory System

### The heart

- 4 chambers: 2 atria (upper) + 2 ventricles (lower).
- Right heart: receives deoxygenated blood → pumps to lungs.
- Left heart: receives oxygenated blood from lungs → pumps to body.
- **Natural pacemaker:** SA node (sinoatrial node), located in right atrium.
- Normal heart rate: 60–100 bpm; normal blood pressure: **120/80 mmHg**.

### Blood composition

| Component        | Amount                           | Function                                                                |
|------------------|----------------------------------|-------------------------------------------------------------------------|
| <b>Plasma</b>    | ~55% of blood                    | Transport medium; contains proteins, hormones                           |
| <b>RBCs</b>      | ~45%; 4–6 million/ $\mu\text{L}$ | O <sub>2</sub> transport via haemoglobin; no nucleus; lifespan 120 days |
| <b>WBCs</b>      | 4,000–11,000/ $\mu\text{L}$      | Immune defence                                                          |
| <b>Platelets</b> | 150,000–450,000/ $\mu\text{L}$   | Clotting                                                                |

#### ⚠ EXAM ALERT

##### Blood group questions — 100% exam reliable:

"Universal donor?" → **O negative (O-)** — no antigens on RBCs, no Rh factor

"Universal recipient?" → **AB positive (AB+)** — no anti-A or anti-B antibodies in plasma

"Who discovered blood groups?" → **Karl Landsteiner** (Nobel Prize 1930)

"Normal Hb: male/female?" → **13–17 / 12–15.5 g/dL**

### WBC types (granulocytes + agranulocytes)

| WBC         | %      | Function                                           |
|-------------|--------|----------------------------------------------------|
| Neutrophils | 60–70% | First responders to bacterial infection            |
| Lymphocytes | 20–40% | T cells (cellular immunity) + B cells (antibodies) |
| Monocytes   | 3–8%   | Largest WBC; become macrophages                    |
| Eosinophils | 1–4%   | Parasites and allergic reactions                   |
| Basophils   | <1%    | Histamine; smallest %                              |

#### × COMMON TRAP

Examiners trick students with: "Biggest WBC?" → **Monocyte**. "Smallest WBC?" → **Lymphocyte**. Many students confuse this with platelets, which are not WBCs at all.

## Chapter D4 — Nervous System

### Divisions

- **Central Nervous System (CNS):** Brain + Spinal cord.
- **Peripheral Nervous System (PNS):** All nerves outside CNS.
- Somatic: voluntary muscles.
- Autonomic: involuntary (sympathetic = fight/flight; parasympathetic = rest/digest).

### Brain regions

| Region                   | Function                                                 |
|--------------------------|----------------------------------------------------------|
| <b>Cerebrum</b>          | Largest part; thinking, memory, voluntary action, speech |
| <b>Cerebellum</b>        | Balance, coordination, posture                           |
| <b>Medulla oblongata</b> | Controls breathing, heart rate, blood pressure           |
| <b>Hypothalamus</b>      | Hunger, thirst, temperature regulation, sleep            |
| <b>Thalamus</b>          | Sensory relay station                                    |

#### ✓ KEY FACT

##### Nervous system exam essentials:

"Largest part of brain?" → **Cerebrum**

"Balance centre?" → **Cerebellum**

"Hunger and thirst centre?" → **Hypothalamus**

"Longest cell in the human body?" → **Neuron**

"Speed of nerve impulse in myelinated fibre?" → **~120 m/s**

Key neurotransmitters: acetylcholine, dopamine, serotonin, GABA, noradrenaline

## Chapter D5 — Endocrine System

| Gland                           | Hormone(s)                        | Function                               | Disorder                                                             |
|---------------------------------|-----------------------------------|----------------------------------------|----------------------------------------------------------------------|
| <b>Pituitary</b><br>(anterior)  | GH, TSH, ACTH, FSH, LH, Prolactin | "Master gland" — controls other glands | GH excess → Gigantism (child), Acromegaly (adult)                    |
| <b>Pituitary</b><br>(posterior) | ADH (Vasopressin), Oxytocin       | Water balance; uterine contraction     | ADH deficiency → Diabetes insipidus                                  |
| <b>Thyroid</b>                  | T3, T4, Calcitonin                | Metabolism; Ca regulation              | Goitre (iodine deficiency); Grave's (hyper); Cretinism (hypo, child) |
| <b>Parathyroid</b>              | PTH                               | Raises blood calcium                   | Tetany (deficiency)                                                  |



| Gland                                       | Hormone(s)                | Function                                        | Disorder                                   |
|---------------------------------------------|---------------------------|-------------------------------------------------|--------------------------------------------|
| <b>Adrenal medulla</b>                      | Adrenaline, Noradrenaline | Fight-or-flight response                        | —                                          |
| <b>Adrenal cortex</b>                       | Cortisol, Aldosterone     | Stress; Na <sup>+</sup> /K <sup>+</sup> balance | Cushing's (excess); Addison's (deficiency) |
| <b>Pancreas (<math>\beta</math>-cells)</b>  | Insulin                   | Lowers blood glucose                            | Diabetes mellitus Type 1 and 2             |
| <b>Pancreas (<math>\alpha</math>-cells)</b> | Glucagon                  | Raises blood glucose                            | —                                          |
| <b>Pineal</b>                               | Melatonin                 | Sleep-wake cycle                                | —                                          |
| <b>Thymus</b>                               | Thymosin                  | T-cell maturation                               | —                                          |

#### ⚠ EXAM ALERT

#### Top endocrine exam questions:

"Master gland?" → **Pituitary**

"Goitre caused by?" → **Iodine deficiency** → **enlarged thyroid**

"Largest endocrine gland?" → **Thyroid**

"Fight-or-flight hormone?" → **Adrenaline (Epinephrine)**

"Insulin deficiency → ?" → **Diabetes mellitus**

"Love hormone / milk ejection?" → **Oxytocin**

## Chapter D6 — Excretory System

- **Kidneys** (2): each contains ~1 million nephrons.
- **Nephron**: functional unit — glomerulus filters blood; Bowman's capsule collects filtrate; tubules reabsorb glucose, water, salts; Loop of Henle concentrates urine.
- Urine: ~96% water + urea + uric acid + creatinine + salts. Yellow colour from **urochrome**.
- Normal output: ~1.5 L/day.
- Other excretory organs: lungs (CO<sub>2</sub>), skin (salts + water via sweat), liver (bile pigments).

## Chapter D7 — Skeletal & Muscular System

### Skeleton

- Adult human: **206 bones** (infant has ~270; many fuse during development).
- **Largest bone**: Femur (thigh).
- **Smallest bone**: Stapes (middle ear).
- **Hardest substance in body**: Tooth enamel.

- Vertebral column: 33 vertebrae → fused to 26 in adults (7 cervical + 12 thoracic + 5 lumbar + sacrum + coccyx).

## Joints

- Ball-and-socket: hip, shoulder (360° movement).
- Hinge: knee, elbow (one plane).
- Pivot: atlas–axis joint (rotation of head).

## Muscles

| Type     | Control     | Structure      | Example                         |
|----------|-------------|----------------|---------------------------------|
| Skeletal | Voluntary   | Striated       | Biceps, quadriceps              |
| Smooth   | Involuntary | Non - striated | Walls of gut, blood vessels     |
| Cardiac  | Involuntary | Striated       | Heart wall — unique combination |

### ✓ KEY FACT

#### Key muscle and bone facts:

"Largest muscle?" → **Gluteus maximus**

"Smallest muscle?" → **Stapedius** (controls stapes in ear)

"Strongest muscle (force/area)?" → **Masseter** (jaw)

"Cardiac muscle is unique because?" → It is **involuntary AND striated** — no other muscle is both.

## PART E — NUTRITION: VITAMINS & MINERALS

### Chapter E1 — Vitamins (Complete Table)

| Vitamin               | Chemical name               | Sources                                 | Deficiency disease                                           |
|-----------------------|-----------------------------|-----------------------------------------|--------------------------------------------------------------|
| <b>A</b>              | Retinol / $\beta$ -carotene | Carrot, liver, milk                     | <b>Night blindness</b> , xerophthalmia                       |
| <b>B<sub>1</sub></b>  | Thiamine                    | Whole grains, pulses                    | <b>Beri - beri</b>                                           |
| <b>B<sub>2</sub></b>  | Riboflavin                  | Milk, eggs                              | Cheilosis, ariboflavinosis                                   |
| <b>B<sub>3</sub></b>  | Niacin                      | Meat, peanuts                           | <b>Pellagra</b> (3 Ds: Dermatitis, Diarrhoea, Dementia)      |
| <b>B<sub>5</sub></b>  | Pantothenic acid            | Widely present                          | Burning feet syndrome                                        |
| <b>B<sub>6</sub></b>  | Pyridoxine                  | Fish, poultry, banana                   | Anaemia                                                      |
| <b>B<sub>7</sub></b>  | Biotin                      | Egg yolk, nuts                          | Hair loss, skin rash                                         |
| <b>B<sub>9</sub></b>  | Folic acid                  | Leafy greens                            | <b>Megaloblastic anaemia</b> ; neural tube defects in foetus |
| <b>B<sub>12</sub></b> | Cyanocobalamin              | Meat, fish, dairy (animal sources only) | <b>Pernicious anaemia</b>                                    |
| <b>C</b>              | Ascorbic acid               | Amla, citrus, guava                     | <b>Scurvy</b> (bleeding gums, weak capillaries)              |
| <b>D</b>              | Calciferol                  | Sunlight, fish liver oil                | <b>Rickets</b> (children), <b>Osteomalacia</b> (adults)      |
| <b>E</b>              | Tocopherol                  | Wheat germ, vegetable oils              | Sterility                                                    |
| <b>K</b>              | Phylloquinone               | Leafy greens, gut bacteria              | <b>Excessive bleeding</b> (clotting failure)                 |

#### ⚠ EXAM ALERT

##### The vitamin exam cheat sheet — learn these cold:

Fat-soluble → **A, D, E, K** (stored in body fat; can accumulate to toxic levels)

Water-soluble → **B-complex + C** (excreted in urine; need daily intake)

Only vitamin NOT found in plants → **B<sub>12</sub>** (vegans must supplement)

Vitamin made by skin + sunlight → **D**

Blood clotting vitamin → **K**

Scurvy → **C** | Night blindness → **A** | Rickets → **D** | Beri-beri → **B<sub>1</sub>**

## Chapter E2 — Essential Minerals

| Mineral           | Function                         | Deficiency                                     |
|-------------------|----------------------------------|------------------------------------------------|
| <b>Iron</b>       | Haemoglobin synthesis            | <b>Anaemia</b>                                 |
| <b>Iodine</b>     | Thyroid hormone synthesis        | <b>Goitre</b>                                  |
| <b>Calcium</b>    | Bones, teeth, muscle contraction | Rickets, osteoporosis                          |
| <b>Phosphorus</b> | Bones, ATP, DNA                  | Rare                                           |
| <b>Sodium</b>     | Fluid balance, nerve impulse     | Hyponatraemia                                  |
| <b>Potassium</b>  | Nerve and muscle function        | Hypokalaemia                                   |
| <b>Fluorine</b>   | Dental enamel hardening          | Dental caries (deficiency); Fluorosis (excess) |
| <b>Zinc</b>       | Enzyme cofactor, immunity        | Poor wound healing, immune deficiency          |

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## PART F — DISEASES & IMMUNITY

### Chapter F1 — Disease Classification

Every exam question about diseases falls into one of four categories: pathogen type, vector, vaccine, or elimination status.

#### Bacterial diseases

| Disease           | Causative agent                    | Vaccine/treatment              |
|-------------------|------------------------------------|--------------------------------|
| Tuberculosis (TB) | <i>Mycobacterium tuberculosis</i>  | <b>BCG</b> vaccine             |
| Cholera           | <i>Vibrio cholerae</i>             | ORS; vaccine                   |
| Typhoid           | <i>Salmonella typhi</i>            | TAB vaccine                    |
| Plague            | <i>Yersinia pestis</i>             | Antibiotics; rat - flea vector |
| Tetanus           | <i>Clostridium tetani</i>          | <b>DPT</b> vaccine             |
| Diphtheria        | <i>Corynebacterium diphtheriae</i> | <b>DPT</b> vaccine             |
| Whooping cough    | <i>Bordetella pertussis</i>        | <b>DPT</b> vaccine             |
| Leprosy           | <i>Mycobacterium leprae</i>        | MDT (multi-drug therapy)       |

#### Viral diseases

| Disease         | Virus             | Notes                                                                  |
|-----------------|-------------------|------------------------------------------------------------------------|
| <b>Smallpox</b> | Variola           | <b>Eradicated 1980</b> — first human disease ever eradicated           |
| <b>Polio</b>    | Poliovirus        | India polio-free since <b>27 March 2014</b> ; OPV (Sabin) + IPV (Salk) |
| Measles         | Paramyxovirus     | MMR vaccine                                                            |
| AIDS            | HIV (retrovirus)  | First reported 1981; reverse transcriptase                             |
| Dengue          | DENV 1-4          | Vector: <b>female Aedes aegypti</b> (day-biter)                        |
| Rabies          | Rabies lyssavirus | Post-exposure vaccine; dog/bat bite                                    |
| COVID-19        | SARS-CoV-2        | Pandemic declared 11 March 2020                                        |
| Hepatitis B     | HBV               | Blood/sexual transmission; vaccine available                           |

## Protozoal & parasitic diseases

| Disease           | Pathogen                     | Vector                           |
|-------------------|------------------------------|----------------------------------|
| <b>Malaria</b>    | <i>Plasmodium</i> spp.       | <b>Female Anopheles</b> mosquito |
| Kala-azar         | <i>Leishmania donovani</i>   | Sand fly                         |
| Sleeping sickness | <i>Trypanosoma</i>           | Tsetse fly                       |
| <b>Filariasis</b> | <i>Wuchereria bancrofti</i>  | <b>Culex</b> mosquito            |
| Amoebiasis        | <i>Entamoeba histolytica</i> | Faecal-oral                      |
| Tapeworm          | <i>Taenia</i> spp.           | Undercooked pork/beef            |

### ⚠ EXAM ALERT

#### Vector mapping — tested in EVERY paper:

Malaria → Female **Anopheles**

Dengue → Female **Aedes aegypti** (also Zika, Chikungunya)

Filariasis → **Culex**

Kala-azar → **Sand fly**

Plague → **Rat flea**

Sleeping sickness → **Tsetse fly**

## Chapter F2 — Immunity

The immune system has two arms. **Innate immunity** is the rapid, non-specific response: skin as a barrier, mucus, and phagocytes that eat any invader without needing prior exposure. **Adaptive immunity** is specific and has memory: B lymphocytes produce antibodies; T lymphocytes kill infected cells directly.

### Antibodies (Immunoglobulins)

| Class      | Key fact                                                             |
|------------|----------------------------------------------------------------------|
| <b>IgG</b> | Most abundant; crosses placenta; main antibody of secondary response |
| <b>IgM</b> | <b>First responder</b> — appears first after infection               |
| <b>IgA</b> | Found in saliva, tears, breast milk (mucosal immunity)               |
| <b>IgE</b> | Allergic reactions; parasites                                        |
| <b>IgD</b> | B-cell activation                                                    |

## Vaccines

| Vaccine                  | Disease                                         |
|--------------------------|-------------------------------------------------|
| BCG                      | Tuberculosis                                    |
| OPV (Sabin) / IPV (Salk) | Polio                                           |
| MMR                      | Measles, Mumps, Rubella                         |
| DPT                      | Diphtheria, Pertussis, Tetanus                  |
| Hepatitis B              | HBV                                             |
| HPV                      | Human papillomavirus (prevents cervical cancer) |

### ✓ KEY FACT

#### Vaccine history points examiners test:

Concept of vaccination pioneered by → **Edward Jenner** (smallpox, 1796, using cowpox)

"First-responder antibody?" → **IgM**

"Most abundant antibody?" → **IgG**

"BCG stands for?" → Bacillus Calmette-Guérin

India's UIP (Universal Immunisation Programme) started → **1985**

# PART G — ECOLOGY & ENVIRONMENT

## Chapter G1 — Ecosystem & Energy Flow

An **ecosystem** is a community of living organisms (biotic) interacting with the non-living environment (abiotic) in a defined area. It can be a pond, a forest, or the entire Earth (biosphere).

### Energy flow rules

- Flow is **one-directional**: Sun → Producers → Consumers → Decomposers. Energy cannot flow backwards.
- **10% rule (Lindeman's law)**: Only 10% of energy at one trophic level transfers to the next. 90% is lost as heat. This limits food chains to 4–5 links.
- **Trophic levels**: T1 = Producers (plants); T2 = Herbivores; T3 = Carnivores; T4 = Top predators. Decomposers operate at all levels.

### Ecological pyramids

| Pyramid type | Always upright?     | Why can it invert?                                         |
|--------------|---------------------|------------------------------------------------------------|
| Numbers      | No                  | One tree supports millions of insects                      |
| Biomass      | No                  | Ocean: tiny phytoplankton support large fish biomass above |
| Energy       | <b>YES — always</b> | Energy always decreases at each level; never inverts       |

#### ✓ KEY FACT

The pyramid of **energy** is ALWAYS upright — this is the most tested ecological fact. The pyramids of numbers and biomass can invert (inverted pyramids are possible). If the question says "which pyramid is always upright?" — the answer is Energy.

## Chapter G2 — Biodiversity

**Three levels of biodiversity**: Genetic diversity, Species diversity, Ecosystem diversity.

**Biodiversity hotspots** (globally 36; defined by >1,500 endemic plants AND >70% habitat lost):

India has **4 hotspots**: 1. Western Ghats (+ Sri Lanka) 2. Eastern Himalayas (Indo-Burma) 3. Indo-Burma (part overlapping NE India) 4. Sundaland (Andaman & Nicobar Islands)

**IUCN Red List categories** (most to least threatened): Extinct (EX) → Extinct in Wild (EW) → Critically Endangered (CR) → Endangered (EN) → Vulnerable (VU) → Near Threatened (NT) → Least Concern (LC)



## Chapter G3 — Indian Wildlife & Conservation

### Major conservation projects

| Project                       | Year | Target                                                 |
|-------------------------------|------|--------------------------------------------------------|
| <b>Project Tiger</b>          | 1973 | Bengal tiger; 55+ reserves                             |
| <b>Project Elephant</b>       | 1992 | Asian elephant                                         |
| <b>Project Crocodile</b>      | 1975 | Mugger, saltwater, gharial                             |
| <b>Cheetah reintroduction</b> | 2022 | Kuno National Park (Namibian + South African cheetahs) |
| <b>Sea Turtle Project</b>     | 1976 | Olive ridley turtles                                   |

### National symbols

- National animal: **Bengal tiger** (since 1973).
- National bird: **Indian peacock** (*Pavo cristatus*).
- National aquatic animal: **Ganges river dolphin** (since 2009).
- National heritage animal: **Elephant** (since 2010).
- National tree: **Indian banyan** (*Ficus benghalensis*).
- National flower: **Lotus** (*Nelumbo nucifera*).

### Protected areas in India

- **National Parks**: ~106 (Jim Corbett was first, established 1936 as Hailey NP).
- **Wildlife Sanctuaries**: ~570.
- **Biosphere Reserves**: 18 (12 under UNESCO MAB programme).
- **Largest National Park**: Hemis (Ladakh).
- **Ramsar Sites** (wetlands): ~85 as of 2026.

#### ⚠ EXAM ALERT

##### Conservation questions — high hit rate:

"India's first National Park?" → **Jim Corbett** (established 1936 as Hailey NP)

"Largest NP in India?" → **Hemis (Ladakh)**

"CITES regulates?" → International trade in endangered species

"Biodiversity Act in India?" → **2002** (follows CBD 1992 Rio)

## APPENDIX A — Human Body Trivia (Memorise Cold)

---

| Fact                    | Value                             |
|-------------------------|-----------------------------------|
| Bones (adult)           | <b>206</b>                        |
| Bones (infant)          | ~270                              |
| Muscles                 | ~640                              |
| Teeth (adult)           | 32                                |
| Teeth (milk/deciduous)  | 20                                |
| Largest bone            | Femur                             |
| Smallest bone           | Stapes (middle ear)               |
| Largest organ           | Skin                              |
| Largest internal organ  | Liver                             |
| Largest gland           | Liver                             |
| Master gland            | Pituitary                         |
| Hardest substance       | Tooth enamel                      |
| Largest muscle          | Gluteus maximus                   |
| Smallest muscle         | Stapedius                         |
| Blood volume (adult)    | 5–6 litres                        |
| Heart beats per day     | ~1,00,000                         |
| Normal body temperature | 37°C (98.6°F)                     |
| Brain weight            | ~1.4 kg                           |
| Neurons in brain        | ~86 billion                       |
| RBC lifespan            | 120 days                          |
| Platelet lifespan       | 7–10 days                         |
| WBC lifespan            | Hours to years (type - dependent) |
| pH of blood             | 7.35–7.45                         |

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## APPENDIX B — Scientific Names (Examiner Favourites)

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| Common name      | Scientific name              |
|------------------|------------------------------|
| Human            | <i>Homo sapiens</i>          |
| Tiger            | <i>Panthera tigris</i>       |
| Lion             | <i>Panthera leo</i>          |
| Leopard          | <i>Panthera pardus</i>       |
| Cheetah          | <i>Acinonyx jubatus</i>      |
| Indian elephant  | <i>Elephas maximus</i>       |
| African elephant | <i>Loxodonta africana</i>    |
| Peacock          | <i>Pavo cristatus</i>        |
| King cobra       | <i>Ophiophagus hannah</i>    |
| Mango            | <i>Mangifera indica</i>      |
| Rice             | <i>Oryza sativa</i>          |
| Wheat            | <i>Triticum aestivum</i>     |
| Maize            | <i>Zea mays</i>              |
| Sugarcane        | <i>Saccharum officinarum</i> |
| Tea              | <i>Camellia sinensis</i>     |
| Lotus            | <i>Nelumbo nucifera</i>      |
| Banyan           | <i>Ficus benghalensis</i>    |
| Neem             | <i>Azadirachta indica</i>    |
| Tulsi            | <i>Ocimum tenuiflorum</i>    |
| Onion            | <i>Allium cepa</i>           |

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## APPENDIX C — Trap-Recognition Card

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| The Trap                            | The Truth                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------|
| Universal donor = O+                | NO → <b>O-</b> (the minus/Rh- is essential)                                 |
| Largest cell = neuron               | NO → Ostrich egg is largest; neuron is longest                              |
| Anopheles = dengue                  | NO → Anopheles = <b>malaria</b> ; Aedes = dengue                            |
| Insulin from $\alpha$ -cells        | NO → <b><math>\beta</math>-cells</b> (Islets of Langerhans) produce insulin |
| Cardiac muscle is voluntary         | NO → involuntary AND striated (unique)                                      |
| Vitamin D deficiency = rickets only | Also <b>osteomalacia</b> in adults                                          |
| Father of genetics = Darwin         | NO → <b>Mendel</b> ; Darwin = natural selection                             |
| Bile produced by gall bladder       | NO → bile PRODUCED by liver; STORED in gall bladder                         |
| Largest endocrine gland = pituitary | NO → <b>Thyroid</b> is largest; pituitary is "master"                       |
| Smallpox eradicated by DDT          | NO → <b>vaccination</b> (Edward Jenner's principle)                         |

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## APPENDIX D — 25-Question Mini-Mock

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**Instructions:** 25 questions, 25 minutes. No negative marking in this self-test.

1. Which organelle is called the "powerhouse of the cell"? (a) Nucleus (b) Ribosome (c) Mitochondria (d) Lysosome
2. Which organelle is called "suicide bags" due to its autolytic function? (a) Mitochondria (b) Lysosomes (c) Golgi body (d) Endoplasmic reticulum
3. Who is called the "Father of Genetics"? (a) Darwin (b) Watson (c) Mendel (d) Lamarck
4. The start codon in mRNA translation is: (a) UAA (b) UAG (c) AUG (d) UGA
5. Which blood group is the universal donor? (a) A+ (b) B- (c) O- (d) AB+
6. The largest bone in the human body is: (a) Tibia (b) Femur (c) Humerus (d) Radius
7. The smallest bone in the human body is: (a) Stapes (b) Malleus (c) Incus (d) Clavicle
8. The largest gland in the human body is: (a) Pancreas (b) Thyroid (c) Liver (d) Adrenal
9. Which gland is called the "master gland"? (a) Thyroid (b) Pineal (c) Adrenal (d) Pituitary
10. The "fight-or-flight" hormone is: (a) Insulin (b) Thyroxine (c) Adrenaline (d) Glucagon
11. Goitre is caused by a deficiency of: (a) Iron (b) Calcium (c) Iodine (d) Zinc
12. Which vitamin is essential for blood clotting? (a) Vitamin A (b) Vitamin C (c) Vitamin D (d) Vitamin K
13. Which vitamin is synthesised by the human body on exposure to sunlight? (a) Vitamin A (b) Vitamin B<sub>12</sub> (c) Vitamin C (d) Vitamin D
14. Which vitamin is found exclusively in animal-source foods? (a) Vitamin C (b) Vitamin B<sub>12</sub> (c) Vitamin D (d) Vitamin K
15. Beri-beri is caused by deficiency of: (a) Vitamin B<sub>1</sub> (Thiamine) (b) Vitamin B<sub>2</sub> (c) Vitamin B<sub>6</sub> (d) Vitamin B<sub>12</sub>
16. The vector of malaria is: (a) Male Anopheles mosquito (b) Female Anopheles mosquito (c) Female Aedes aegypti (d) Culex mosquito
17. The vector of dengue fever is: (a) Male Anopheles mosquito (b) Culex mosquito (c) Female Aedes aegypti (d) Female Anopheles mosquito
18. The vector of filariasis is: (a) Aedes aegypti (b) Anopheles (c) Culex (d) Sandfly

19. The first disease to be eradicated globally through vaccination was: (a) Polio (b) Smallpox (c) Tuberculosis (d) Measles
20. India was officially declared polio-free in: (a) 2012 (b) 2013 (c) 2014 (d) 2015
21. The balance and coordination centre of the brain is: (a) Cerebrum (b) Medulla oblongata (c) Hypothalamus (d) Cerebellum
22. The largest part of the human brain is: (a) Cerebellum (b) Cerebrum (c) Medulla (d) Thalamus
23. The pyramid of energy in an ecosystem is: (a) Always inverted (b) Always upright (c) Upright only in aquatic ecosystems (d) Varies by season
24. The 10% energy rule in ecology means: (a) 10% of energy is lost to respiration; 90% passes on (b) Only 10% of energy at one trophic level transfers to the next (c) 10% of energy is stored as biomass permanently (d) 10% of producers are consumed by primary consumers
25. How many biodiversity hotspots does India have? (a) Two (b) Three (c) Four (d) Five
- 

**Answer key:** 1-c | 2-b | 3-c | 4-c | 5-c | 6-b | 7-a | 8-c | 9-d | 10-c | 11-c | 12-d | 13-d | 14-b | 15-a | 16-b | 17-c | 18-c | 19-b | 20-c | 21-d | 22-b | 23-b | 24-b | 25-c

**Scoring:** 23–25 = Excellent | 18–22 = Good | Below 18 = revise weak chapters before your mock series.

**⚠ EXAM ALERT**

**\*\*Error-log rule:\*\*** For every question you got wrong, write one sentence explaining what you missed. Retest yourself on only those questions 3 days later — without looking at the solution. This single habit doubles what the mock teaches you.